HW 4 - Additional 2

a) Given: Required load to crack macadamia nut (222.18 kg)

Diameter of macadamia nut (20 mm → goodness.com.au)

Grip strength (~40 kg)

Find: Dimensions to make nutcracker feasible

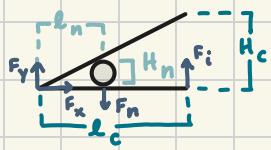
Plan: 1) FBD

2) Moment balance about joint

3) Find ℓ ratio

4) Find H ratio

Solve:



$$\Sigma M = \ell_c \times F_i - \ell_n \times F_n = 0$$

$$\frac{\ell_c}{\ell_n} = \frac{F_n}{F_i} = \frac{222.18\text{ kg}}{40\text{ kg}} = 5.55$$

$$\downarrow$$

$$H_c = 5.55\, H_n$$

$$H_c = 5.55(2\text{ cm}) = 11\text{ cm}$$

Average nutcracker ℓ_n is around 4 cm, so $\ell_c = 22.22$ cm

↳ The nutcracker should be around 22 cm long with nut ~4 cm from the joint (assuming full force ~40 kg is applied). The arms will be ~11 cm apart at this point

b) If the arms are around 11 cm apart when fully closed on the nut, the design is likely not practical. The average hand size is only a few cm longer than this at ~18 cm (healthline.com) and the optimal grip size for a tool is <u>much</u> smaller at ~20% of hand length. This design is too large when considering the wider open conformation.